THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Distance is a: 1

(a) unit **(b)** force **(c)** vector **(d)** scalar

2. The area under a v-t graph gives: 1

(a) force **(b)** acceleration **(c)** displacement **(d)** speed

3. A car travels 100 km in 2 hours. Its average speed is: 1

(a) 200 km/h **(b)** 100 km/h **(c)** 2 km/h **(d)** 50 km/h

4. Displacement is a: 1

(a) vector **(b)** unit **(c)** scalar **(d)** speed

5. The slope of a v-t graph gives: 1

(a) acceleration **(b)** distance **(c)** speed **(d)** force

6. A man walks 10 m in 5 s. His speed is: 1

(a) 5 m/s **(b)** 2 m/s **(c)** 0.5 m/s **(d)** 50 m/s

7. Speed = 1

(a) mass/volume **(b)** distance x time **(c)** time/distance **(d)** distance/time

8. A runner completes one lap of a circular track. The displacement is: 1

(a) double **(b)** equal to distance
(c) zero **(d)** half

9. Velocity changes from 0 to 10 m/s in 2 s. Acceleration: 1

(a) 2 m/s squared **(b)** 10 m/s squared **(c)** 5 m/s squared **(d)** 20 m/s squared

10. Velocity = 1

(a) mass x time **(b)** time/displacement
(c) distance x time **(d)** displacement/time

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define distance. 2

12. Define acceleration. 2

13. Write the second equation of motion. 2

14. Define displacement. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. List three differences between distance and displacement. 3

16. Convert 108 km/h to m/s. 3

17. Name three scalars and three vectors. 3

OR

List three differences between speed and velocity. 3

18. Convert 20 m/s to km/h. 3

OR

A man walks 4 m east and 3 m north. Find the displacement. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain distance, displacement, speed, velocity and acceleration. 5

OR

Find the average speed: 60 km in 2 h and 40 km in 1 h. 5

20. Derive v = u + at. 5

OR

A car at 20 m/s brakes with retardation of 4 m/s squared. Find the time and distance to stop. 5

END OF QUESTION PAPER Total: 40 Marks

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